

Estrogen levels in nipple aspirate fluid and serum during a randomized soy trial

Background Based on the hypothesized protective effect, we examined the effect of soy foods on estrogens in nipple aspirate fluid (NAF) and serum, possible indicators of breast cancer risk. **Methods** In a cross-over design, we randomized 96 women who produced ≥ 10 μL NAF to a high- or low-soy diet for 6-months. During the high-soy diet, participants consumed 2 soy servings of soy milk, tofu, or soy nuts (approximately 50 mg of isoflavones/day); during the low-soy diet, they maintained their usual diet. Six NAF samples were obtained using a FirstCyte© Aspirator. Estradiol (E2) and estrone sulfate (E1S) were assessed in NAF and estrone (E1) in serum only using highly sensitive radioimmunoassays. Mixed-effects regression models accounting for repeated measures and left-censoring limits were applied. **Results** Mean E2 and E1S were lower during the high-soy than the low-soy diet (113 vs. 313 pg/mL and 46 vs. 68 ng/mL, respectively) without reaching significance ($p=0.07$); the interaction between group and diet and was not significant. There was no effect of the soy treatment on serum E2 ($p=0.76$), E1 ($p=0.86$), or E1S ($p=0.56$). Within individuals, NAF and serum levels of E2 ($r_s=0.37$; $p<0.001$) but not E1S ($r_s=0.004$; $p=0.97$) were correlated. E2 and E1S in NAF and serum were strongly associated ($r_s=0.78$ and $r_s=0.48$; $p<0.001$). **Conclusions** Soy foods in amounts consumed by Asians did not significantly modify estrogen levels in NAF and serum. **Impact** The trend towards lower estrogens in NAF during the high-soy diet counters concerns about adverse effects of soy foods on breast cancer risk.